# 使用 Jarvis， Cloud Sync 和 Nemo 构建虚拟助手

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# 概述

本节详细介绍了虚拟零售助理的实施。

"接下来： JarVis 部署"

## JarVis 部署

您可以注册 "JARVIS 早期访问计划" 访问 NVIDIA GPU Cloud （ NGC ） 上的 JarVis 容器。从 NVIDIA 收到凭据后，您可以使用以下步骤部署 JarVis ：

1. 登录到 NGC 。
2. 在 NGC 上设置您的组织： `ea-2-JarVis` 。
3. 找到 JarVis EA v0.2 资产： JarVis containers are in `Private Registry` > `Organization Containers` 。
4. 选择 JarVis ： 导航到 `Model Scripts` ，然后单击 `JarVis Quick Start`
5. 验证所有资产是否均正常工作。
6. 查找用于构建您自己的应用程序的文档： PDF 可在 `Model Scripts` > `JarVis Documentation` > `File Browser` 中找到。

"接下来：针对零售用例自定义状态和流程"

## 针对零售用例自定义状态和流程

您可以针对特定使用情形自定义对话框管理器的状态和流。在我们的零售示例中，我们提供了以下四个 YAML 文件，用于根据不同意向指导对话。

查看以下文件名列表以及每个文件的问题描述：

- `main_flow.yml` ： 定义主要对话流和状态，并在必要时将此流定向到其他三个 YAML 文件。
- `retail _flow.yml` ： 包含与零售或感兴趣点问题相关的状态。系统会提供最近商店的信息或给定商品的价格。
- `weather flow.yml` ： 包含与天气问题相关的状态。如果无法确定位置，系统会询问一个跟进问题以进行澄清。
- `error_flow.yml` ： 处理用户意向不属于上述三个 YAML 文件的情况。显示错误消息后，系统会重新路由到接受用户问题。以下各节包含这些 YAML 文件的详细定义。

### main_flow.yml

```
name: JarvisRetail
intent_transitions:
  jarvis_error: error
  price_check: retail_price_check
```

```
  inventory_check: retail_inventory_check
  store_location: retail_store_location
  weather.weather: weather
  weather.temperature: temperature
  weather.sunny: sunny
  weather.cloudy: cloudy
  weather.snow: snow
  weather.rainfall: rain
  weather.snow_yes_no: snowfall
  weather.rainfall_yes_no: rainfall
  weather.temperature_yes_no: tempyesno
  weather.humidity: humidity
  weather.humidity_yes_no: humidity
  navigation.startnavigationpoi: retail # Transitions should be context
and slot based. Redirecting for now.
  navigation.geteta: retail
  navigation.showdirection: retail
  navigation.showmappoi: idk_what_you_talkin_about
  nomatch.none: idk_what_you_talkin_about
states:
  init:
    type: message_text
    properties:
      text: "Hi, welcome to NARA retail and weather service. How can I
help you?"
  input_intent:
    type: input_context
    properties:
      nlp_type: jarvis
      entities:
        intent: dontcare
# This state is executed if the intent was not understood
  dont_get_the intent:
    type: message_text_random
    properties:
      responses:
        - "Sorry I didn't get that! Please come again."
        - "I beg your pardon! Say that again?"
        - "Are we talking about weather? What would you like to know?"
        - "Sorry I know only about the weather"
        - "You can ask me about the weather, the rainfall, the
temperature, I don't know much more"
      delay: 0
    transitions:
      next_state: input_intent
  idk_what_you_talkin_about:
```

```
    type: message_text_random
    properties:
      responses:
        - "Sorry I didn't get that! Please come again."
        - "I beg your pardon! Say that again?"
        - "Are we talking about retail or weather? What would you like to
know?"
        - "Sorry I know only about retail and the weather"
        - "You can ask me about retail information or the weather, the
rainfall, the temperature. I don't know much more."
      delay: 0
    transitions:
      next_state: input_intent
  error:
    type: change_context
    properties:
        update_keys:
           intent: 'error'
    transitions:
        flow: error_flow
  retail_inventory_check:
    type: change_context
    properties:
        update_keys:
           intent: 'retail_inventory_check'
    transitions:
        flow: retail_flow
  retail_price_check:
    type: change_context
    properties:
        update_keys:
           intent: 'check_item_price'
    transitions:
        flow: retail_flow
  retail_store_location:
    type: change_context
    properties:
        update_keys:
           intent: 'find_the_store'
    transitions:
        flow: retail_flow
  weather:
    type: change_context
    properties:
        update_keys:
           intent: 'weather'
```

```
    transitions:
        flow: weather_flow
  temperature:
    type: change_context
    properties:
        update_keys:
           intent: 'temperature'
    transitions:
        flow: weather_flow
  rainfall:
    type: change_context
    properties:
        update_keys:
           intent: 'rainfall'
    transitions:
        flow: weather_flow
  sunny:
    type: change_context
    properties:
        update_keys:
           intent: 'sunny'
    transitions:
        flow: weather_flow
  cloudy:
    type: change_context
    properties:
        update_keys:
           intent: 'cloudy'
    transitions:
        flow: weather_flow
  snow:
    type: change_context
    properties:
        update_keys:
           intent: 'snow'
    transitions:
        flow: weather_flow
  rain:
    type: change_context
    properties:
        update_keys:
           intent: 'rain'
    transitions:
        flow: weather_flow
  snowfall:
      type: change_context
```

```
      properties:
          update_keys:
             intent: 'snowfall'
      transitions:
          flow: weather_flow
  tempyesno:
      type: change_context
      properties:
          update_keys:
             intent: 'tempyesno'
      transitions:
          flow: weather_flow
  humidity:
      type: change_context
      properties:
          update_keys:
             intent: 'humidity'
      transitions:
          flow: weather_flow
  end_state:
    type: reset
    transitions:
      next_state: init
```

### Retail， flow.yml

```
name: retail_flow
states:
  store_location:
    type: conditional_exists
    properties:
      key: '{{location}}'
    transitions:
      exists: retail_state
      notexists: ask_retail_location
  retail_state:
    type: Retail
    properties:
    transitions:
      next_state: output_retail
  output_retail:
      type: message_text
      properties:
        text: '{{retail_status}}'
      transitions:
```

```
        next_state: input_intent
  ask_retail_location:
    type: message_text
    properties:
      text: "For which location? I can find the closest store near you."
    transitions:
      next_state: input_retail_location
  input_retail_location:
    type: input_user
    properties:
      nlp_type: jarvis
      entities:
        slot: location
      require_match: true
    transitions:
      match: retail_state
      notmatch: check_retail_jarvis_error
  output_retail_acknowledge:
    type: message_text_random
    properties:
      responses:
        - 'ok in {{location}}'
        - 'the store in {{location}}'
        - 'I always wanted to shop in {{location}}'
      delay: 0
    transitions:
      next_state: retail_state
  output_retail_notlocation:
    type: message_text
    properties:
      text: "I did not understand the location. Can you please repeat?"
    transitions:
      next_state: input_intent
  check_rerail_jarvis_error:
    type: conditional_exists
    properties:
      key: '{{jarvis_error}}'
    transitions:
      exists: show_retail_jarvis_api_error
      notexists: output_retail_notlocation
  show_retail_jarvis_api_error:
    type: message_text
    properties:
      text: "I am having troubled understanding right now. Come again on
that?"
    transitions:
```

```
      next_state: input_intent
```

### weather flow.yml

```
name: weather_flow
states:
  check_weather_location:
    type: conditional_exists
    properties:
      key: '{{location}}'
    transitions:
      exists: weather_state
      notexists: ask_weather_location
  weather_state:
    type: Weather
    properties:
    transitions:
      next_state: output_weather
  output_weather:
      type: message_text
      properties:
        text: '{{weather_status}}'
      transitions:
        next_state: input_intent
  ask_weather_location:
    type: message_text
    properties:
      text: "For which location?"
    transitions:
      next_state: input_weather_location
  input_weather_location:
    type: input_user
    properties:
      nlp_type: jarvis
      entities:
        slot: location
      require_match: true
    transitions:
      match: weather_state
      notmatch: check_jarvis_error
  output_weather_acknowledge:
    type: message_text_random
    properties:
      responses:
        - 'ok in {{location}}'
```

```
        - 'the weather in {{location}}'
        - 'I always wanted to go in {{location}}'
      delay: 0
    transitions:
      next_state: weather_state
  output_weather_notlocation:
    type: message_text
    properties:
      text: "I did not understand the location, can you please repeat?"
    transitions:
      next_state: input_intent
  check_jarvis_error:
    type: conditional_exists
    properties:
      key: '{{jarvis_error}}'
    transitions:
      exists: show_jarvis_api_error
      notexists: output_weather_notlocation
  show_jarvis_api_error:
    type: message_text
    properties:
      text: "I am having troubled understanding right now. Come again on
that, else check jarvis services?"
    transitions:
      next_state: input_intent
```

### error_flow.yml

```
name: error_flow
states:
  error_state:
    type: message_text_random
    properties:
      responses:
        - "Sorry I didn't get that!"
        - "Are we talking about retail or weather? What would you like to
know?"
        - "Sorry I know only about retail information or the weather"
        - "You can ask me about retail information or the weather, the
rainfall, the temperature. I don't know much more"
        - "Let's talk about retail or the weather!"
      delay: 0
    transitions:
      next_state: input_intent
```

"下一步：连接到第三方 API 作为实施引擎"

# 以履行引擎的形式连接到第三方 API

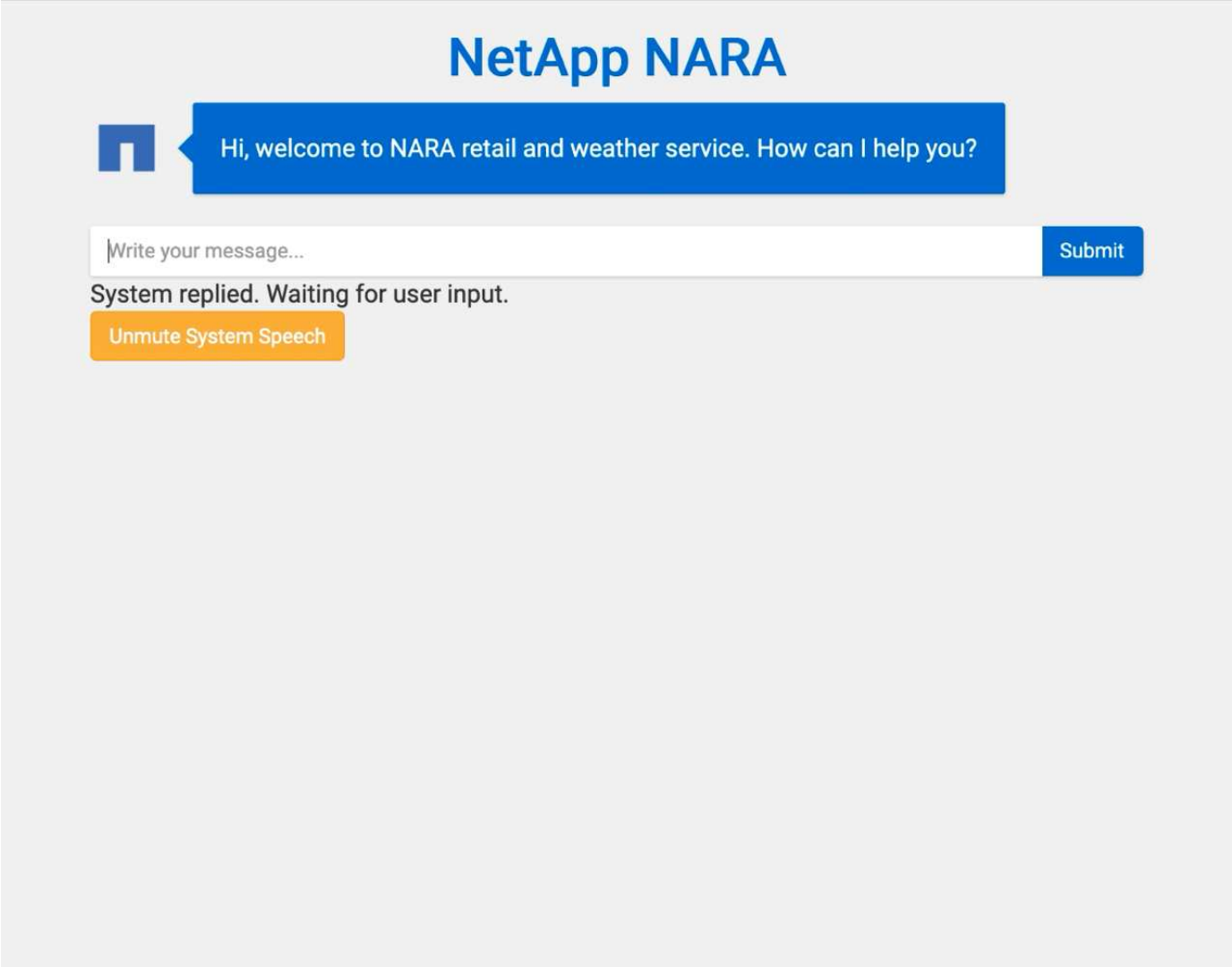
我们将以下第三方 API 作为履行引擎连接到问题解答问题：

- "WeatherStack API"：返回给定位置的天气，温度，雨和雪。
- "Yelp Fusion API"：返回给定位置中最接近的存储信息。
- "eBay Python SDK"：返回给定项目的价格。

"接下来： NetApp 零售助理演示"

# NetApp 零售助理演示

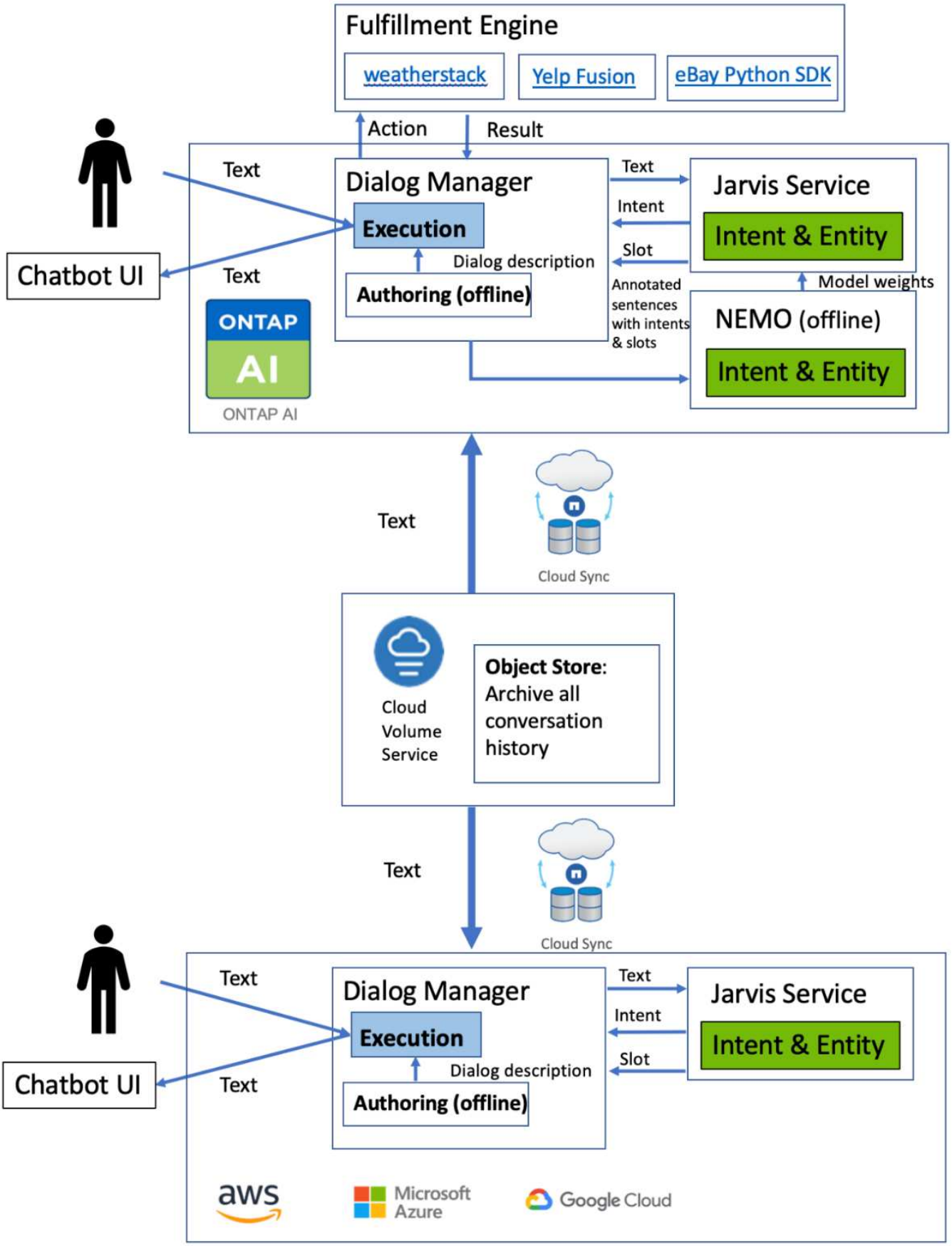
我们录制了 NetApp 零售助理（ Nara ）的演示视频。单击 "此链接。" 打开下图并播放视频演示。




"下一步：使用 NetApp Cloud Sync 归档对话历史记录"

# 使用 NetApp Cloud Sync 归档对话历史记录

通过每天将对话历史记录转储到 CSV 文件中一次，我们可以利用 Cloud Sync 将日志文件下载到本地存储中。下图显示了在内部和公有云中部署 JarVis ，同时使用 Cloud Sync 发送 Nemo 培训对话历史记录的架构。有关 Nemo 培训的详细信息，请参见一节 "使用 Nemo 培训扩展意向模型"。




"接下来：使用 Nemo 培训扩展意向模型"

# 使用 Nemo 培训扩展意向模型

NVIDIA Nemo 是由 NVIDIA 构建的一个工具包，用于创建对话式 AI 应用程序。该工具包包含一系列针对 ASL ， NLP 和 TTS- 的预培训模块，使研究人员和数据科学家能够轻松构建复杂的神经网络架构，并更加专注于设计自己的应用程序。

如上例所示， Nara 只能处理有限类型的问题。这是因为经过预先培训的 NLP 模型只会对这些类型的问题进行训练。如果我们希望 Nara 能够处理更广泛的问题，我们需要使用自己的数据集对其进行重新训练。因此，我们将在此演示如何使用 Nemo 扩展 NLP 模型以满足要求。我们首先将从 Nara 收集的日志转换为 Nemo 的格式，然后训练数据集以增强 NLP 模型。

## 型号

我们的目标是使 Nara 能够根据用户首选项对项目进行排序。例如，我们可能会要求 Nara 推荐排名最高的寿司店，也可能希望 Nara 寻找价格最低的 jeans 。为此，我们使用 Nemo 中提供的意向检测和插槽填充模型作为我们的培训模型。通过此模型， Nara 可以了解搜索首选项的意图。

## 数据准备

为了训练模型，我们会收集此类问题的数据集，并将其转换为 Nemo 格式。我们在此处列出了用于训练模型的文件。

### dict.intents.csv

此文件列出了我们希望 Nemo 了解的所有意向。此处，我们有两个主要意向，一个意图仅用于对不符合任何主要意向的问题进行分类。

```
price_check
find_the_store
unknown
```

### dict.slots.csv

此文件列出了我们可以在培训问题上标记的所有插槽。

```
B-store.type
B-store.name
B-store.status
B-store.hour.start
B-store.hour.end
B-store.hour.day
B-item.type
B-item.name
B-item.color
B-item.size
B-item.quantity
B-location
```

```
B-cost.high
B-cost.average
B-cost.low
B-time.period_of_time
B-rating.high
B-rating.average
B-rating.low
B-interrogative.location
B-interrogative.manner
B-interrogative.time
B-interrogative.personal
B-interrogative
B-verb
B-article
I-store.type
I-store.name
I-store.status
I-store.hour.start
I-store.hour.end
I-store.hour.day
I-item.type
I-item.name
I-item.color
I-item.size
I-item.quantity
I-location
I-cost.high
I-cost.average
I-cost.low
I-time.period_of_time
I-rating.high
I-rating.average
I-rating.low
I-interrogative.location
I-interrogative.manner
I-interrogative.time
I-interrogative.personal
I-interrogative
I-verb
I-article
O
```

**Traine.tsv**

这是主要的培训数据集。每行都以文件 dict.intent.csv 中列出的意图类别后面的问题开头。此标签将从零开始枚举。

**Train_slots.tsv**

```
20 46 24 25 6 32 6
52 52 24 6
23 52 14 40 52 25 6 32 6
…
```

## 训练模型

```
docker pull nvcr.io/nvidia/nemo:v0.10
```

然后，我们将使用以下命令启动此容器。在此命令中，我们会将容器限制为使用单个 GPU （ GPU ID = 1 ），因为这是一项轻型训练练习。此外，我们还会将本地工作空间 /workstore/nemo/ 映射到容器 /nemo 中的文件夹。

```
NV_GPU='1' docker run --runtime=nvidia -it --shm-size=16g \
                        --network=host --ulimit memlock=-1 --ulimit
stack=67108864 \
                        -v /workspace/nemo:/nemo\
                        --rm nvcr.io/nvidia/nemo:v0.10
```

在容器中，如果要从最初的预先培训的 Bert 模型开始，我们可以使用以下命令启动培训操作步骤。data_dir 是用于设置训练数据路径的参数。work_dir 用于配置检查点文件的存储位置。

```
cd examples/nlp/intent_detection_slot_tagging/
python joint_intent_slot_with_bert.py \
    --data_dir /nemo/training_data\
    --work_dir /nemo/log
```

如果我们有新的培训数据集并希望改进先前的模型，则可以使用以下命令从停止的位置继续操作。checkpoint_dir 获取上一个检查点文件夹的路径。

```
cd examples/nlp/intent_detection_slot_tagging/
python joint_intent_slot_infer.py \
    --data_dir /nemo/training_data \
    --checkpoint_dir /nemo/log/2020-05-04_18-34-20/checkpoints/ \
    --eval_file_prefix test
```

## 推理模型

我们需要在经过一定次数的时间之后验证经过训练的模型的性能。使用以下命令，我们可以逐个测试查询。例如，在此命令中，我们希望检查我们的模型是否能够正确识别查询的目的 在哪里可以获得最好的意大利面 。

```
cd examples/nlp/intent_detection_slot_tagging/
python joint_intent_slot_infer_b1.py \
--checkpoint_dir /nemo/log/2020-05-29_23-50-58/checkpoints/ \
--query "where can i get the best pasta" \
--data_dir /nemo/training_data/ \
--num_epochs=50
```

然后，以下是推理的输出。在输出中，我们可以看到经过培训的模型可以正确预测 DETAINT_the_store 的意向，并返回我们感兴趣的关键字。通过这些关键字，我们可以使 Nara 搜索用户所需内容并进行更精确的搜索。

```
[NeMo I 2020-05-30 00:06:54 actions:728] Evaluating batch 0 out of 1
[NeMo I 2020-05-30 00:06:55 inference_utils:34] Query: where can i get the
best pasta
[NeMo I 2020-05-30 00:06:55 inference_utils:36] Predicted intent:       1
find_the_store
[NeMo I 2020-05-30 00:06:55 inference_utils:50] where   B-
interrogative.location
[NeMo I 2020-05-30 00:06:55 inference_utils:50] can     O
[NeMo I 2020-05-30 00:06:55 inference_utils:50] i       O
[NeMo I 2020-05-30 00:06:55 inference_utils:50] get     B-verb
[NeMo I 2020-05-30 00:06:55 inference_utils:50] the     B-article
[NeMo I 2020-05-30 00:06:55 inference_utils:50] best    B-rating.high
[NeMo I 2020-05-30 00:06:55 inference_utils:50] pasta   B-item.type
```

"接下来：总结。"

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